Conditional Probability

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1 Setup

Throughout, let $(\Omega, \mathcal{F}, P)$ be a probability space. We use $A, B, \dots$ for elements of $\mathcal{F}$ ("events").

2 Definitions

Definition 1 (Conditional probability)**.** Let $B \in \mathcal{F}$ with $P(B) > 0$. For any $A \in \mathcal{F}$, the *conditional probability of A given B* is

$$P(A \mid B) \;:=\; \frac{P(A \cap B)}{P(B)}.$$

Proposition 2 ($P(\cdot \mid B)$ is a probability measure)**.** *Fix B with $P(B) > 0$. The set function $Q : \mathcal{F} \to [0,1]$ given by $Q(A) = P(A \mid B)$ is a probability measure on $(\Omega, \mathcal{F})$.*

Proof. Non-negativity: $P(A \cap B) \geq 0$ and $P(B) > 0$, so $Q(A) \geq 0$. Normalization: $Q(\Omega) = P(\Omega \cap B)/P(B) = P(B)/P(B) = 1$. Countable additivity: if $\{A_n\}$ are pairwise disjoint, then so are $\{A_n \cap B\}$, so

$$Q\left(\bigcup_n A_n\right) = \frac{P(\bigcup_n (A_n \cap B))}{P(B)} = \frac{\sum_n P(A_n \cap B)}{P(B)} = \sum_n Q(A_n). \qquad \square$$

Definition 3 (Independence)**.** Events A, B are *independent* iff $P(A \cap B) = P(A)P(B)$. Equivalently, when $P(B) > 0$, iff $P(A \mid B) = P(A)$.

3 Total Probability and Bayes

Lemma 4 (Law of total probability)**.** *Let $\{B_i\}_{i=1}^n$ be a finite measurable partition of Ω with $P(B_i) > 0$ for all i. For any $A \in \mathcal{F}$,*

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i).$$

Proof. Since $\{B_i\}$ partitions Ω, the sets $\{A \cap B_i\}$ partition A. By finite additivity of P and the definition of conditional probability,

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i) P(B_i). \qquad \square$$

Theorem 5 (Bayes)**.** *Let $A, B \in \mathcal{F}$ with $P(A) > 0$ and $P(B) > 0$. Then*

$$P(A \mid B) \;=\; \frac{P(B \mid A)\, P(A)}{P(B)}.$$

Moreover, if $\{A_i\}_{i=1}^n$ is a measurable partition of Ω with each $P(A_i) > 0$, then for any j,

$$P(A_j \mid B) \;=\; \frac{P(B \mid A_j)P(A_j)}{\sum_{i=1}^n P(B \mid A_i)P(A_i)}.$$

Proof. By definition,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

The second identity yields $P(A \cap B) = P(B \mid A)\,P(A)$. Substituting into the first gives

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)},$$

which is the first claim. For the second, apply the law of total probability with the partition $\{A_i\}$ to expand $P(B) = \sum_i P(B \mid A_i)P(A_i)$ in the denominator. □

4 Chain rule

Proposition 6 (Multiplication / chain rule)**.** *Let $A_1, \ldots, A_n \in \mathcal{F}$ with $P(A_1 \cap \cdots \cap A_{n-1}) > 0$. Then*

$$P(A_1 \cap \cdots \cap A_n) \;=\; P(A_1) \prod_{k=2}^{n} P(A_k \mid A_1 \cap \cdots \cap A_{k-1}).$$

Proof. Induct on n. The base $n = 2$ is the definition rearranged: $P(A_1 \cap A_2) = P(A_2 \mid A_1)P(A_1)$. For the inductive step, apply the base case to $A_1 \cap \cdots \cap A_{n-1}$ and A_n. □

5 Example

Example 7 (Monty Hall)**.** Three doors; a car uniformly hidden behind one. The contestant picks door 1; the host (knowing the layout) opens a goat door from $\{2, 3\}$, breaking ties uniformly. Conditional on the host opening door 3, Bayes gives $P(\text{car at 2} \mid \text{open 3}) = 2/3$. Switching is optimal.